Practice Exam 3 v2 - MAC 1105C Osprey1st College Algebra

Summer 2024

Name: ______________________

Please show your work. Answers with no work may not give you any credit.

1. Solve the following equations.

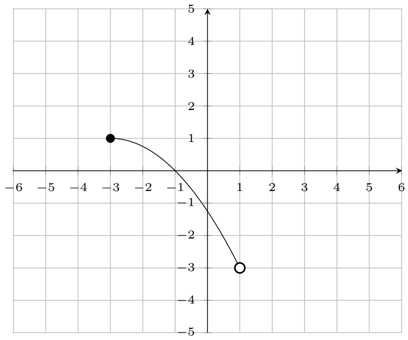
   (a) $\sqrt{24x+25}-5=-2+2x$

   $+5 \quad +5$

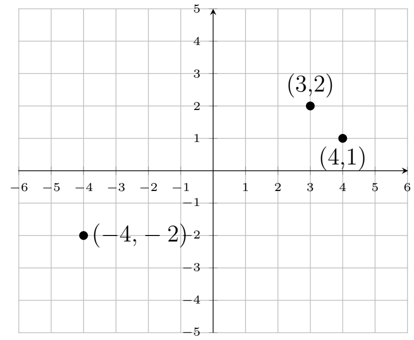
   $\sqrt{24x+25} =$

   (b) $\dfrac{2x+1}{x-2}+\dfrac{4}{x}=\dfrac{-8}{x^2-2x}$

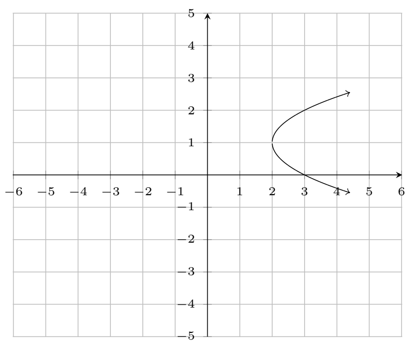
2. Give the domain and range of each relation. Determine whether the relation defines a function.

(a) 

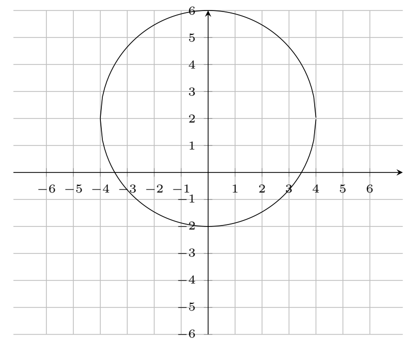
Function (yes/no): ____________
Domain: ____________
Range: ____________

(c) 

Function (yes/no): ____________
Domain: ____________
Range: ____________

(b) 

Function (yes/no): ____________
Domain: ____________
Range: ____________

(d) 

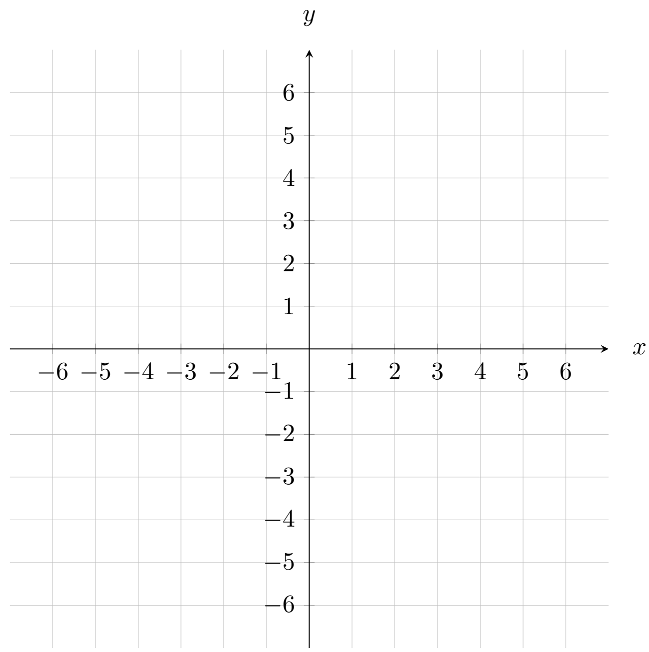
Function (yes/no): ____________
Domain: ____________
Range: ____________

3. The function f is defined by $f(x) = 2x - 5$. Find $f(x+2)$.

4. For the real-valued functions $f(x) = x^2 + 2$ and $g(x) = \sqrt{x+3}$, find the composition $f \circ g$ and specify its domain using interval notation.

5. Graph the following lines. For each line, determine its slope and y-intercept.

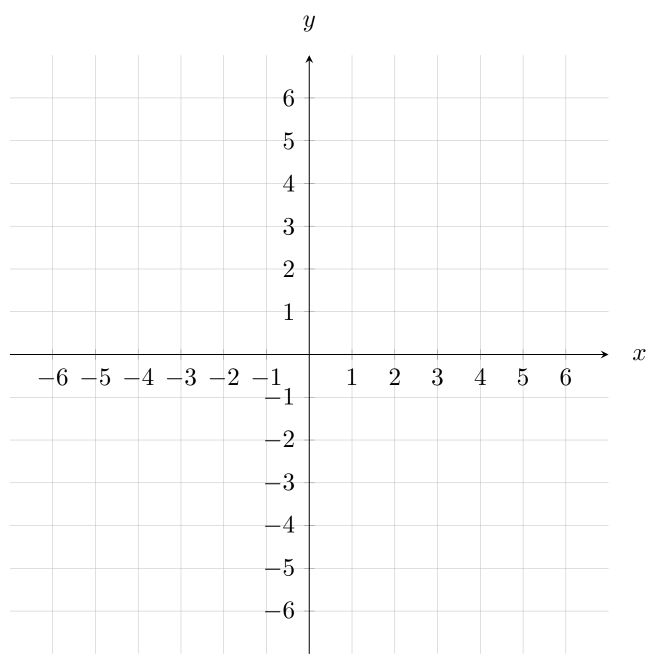
(a) $y = -\frac{4}{7}x + 3$



Slope: ______________

Y-Intercept: ______________

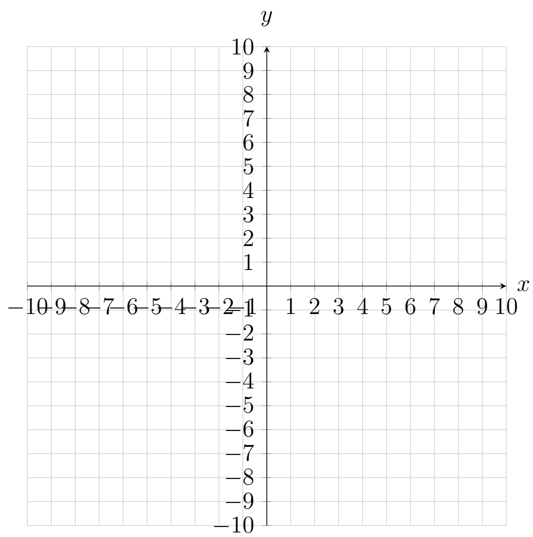
(b) $y = 0$



Slope: ______________

Y-Intercept: ______________

6. Graph the quadratic function $f(x) = -2x^2 + 4x - 1$. Plot five points on the parabola: the vertex, two points to the left of the vertex, and two points to the right of the vertex



7. Solve the following quadratic inequality. Write your answers in interval notations.

$$x^2 + 2x \leq 63$$

8. For the given one-to-one function $f(x) = 2x^5 - 5$ find an equation of the inverse function, $f^{-1}(x)$.

9. Solve the following system.

$$\begin{aligned} 2x - y &= 7 \\ x + 5y &= 9 \end{aligned}$$

10. Solve the equations.

(a) $8^{3x} = 4^{x-3}$

(b) $\log_6 x = 2 - \log_6(x + 9)$

11. Suppose \$4000 is invested in an account paying 3% interest per year. How long will it take \$4000 to double, if interest is compounded continuously? Give the exact answer. (Continuous compounding formula: $A = Pe^{rt}$).